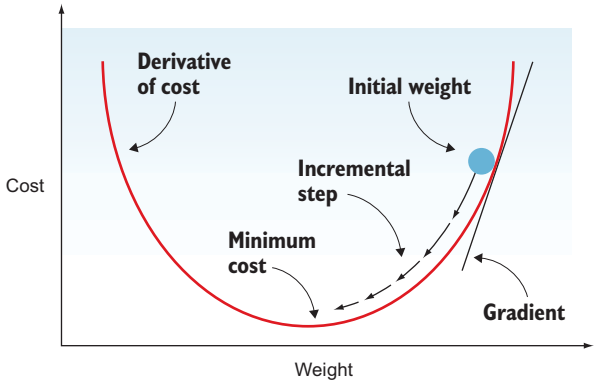

Figure 2.30 Gradient descent takes incremental steps to descend the error function.

HOW DOES GRADIENT DESCENT WORK?

To visualize how gradient descent works, let's plot the error function in a 3D graph (figure 2.31) and go through the process step by step. The random initial weight (starting weight) is at point A, and our goal is to descend this error mountain to the goal w_1 and w_2 weight values, which produce the minimum error value. The way we do that is by taking a series of steps *down* the curve until we get the minimum error. In order to descend the error mountain, we need to determine two things for each step:

- The step direction (gradient)
- The step size (learning rate)

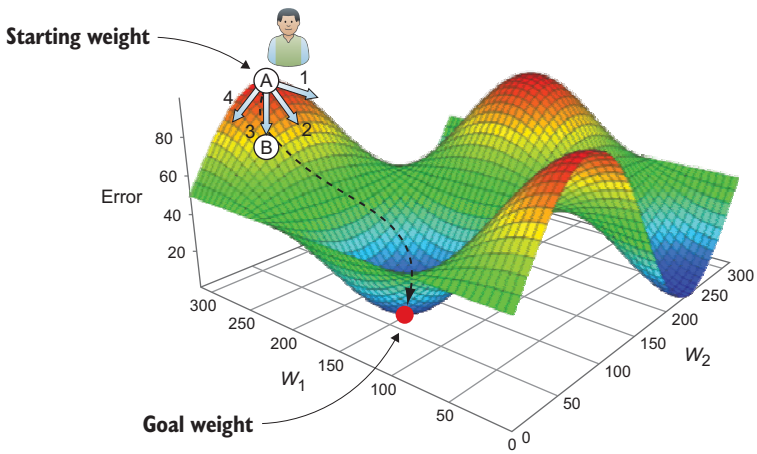

Figure 2.31 The random initial weight (starting weight) is at point A. We descend the error mountain to the w_1 and w_2 weight values that produce the minimum error value.